

Education

- 2010.9–2013.6 Software School of Tsinghua University, Master of Software Engineering
2006.9–2010.6 Beihang University, Bachelor of Computer Science and Technology

Working Experiences

- 2017.12 - Present ByteDance, Senior Engineer, Level 3.1, Performance Rating: M+
2016.11 - 2017.12 Huawei 2012 Distributed Lab, Senior Engineer A (Distributed Database and System), Level 17, Performance Rating: B
2015.10 - 2016.11 Meituan Cloud Computing Object Storage Group, Senior Engineer(Distributed Storage), Level T3.1, Performance Rating: A
2014.11 - 2015.9 Sohu Big Data Center, Recommendation System and Precision Advertising Team, Senior Development Engineer, Level L3.1, Performance Rating: AA
2013.7 - 2014.10 Alipay Beijing OceanBase Team, Development Engineer (Distributed Database Development), Level P5, Performance Rating: 3.5

Language Skills

- ▶ Fluent in English, skilled in documentation reading and writing.CET4(548),CET6(549),

Patents and Paper

- 2016 Patent: Zhangshuai;ChenXiaohui;Yinyin;Likai; Data storage method and apparatus, and data recovery method and apparatus [CN106201764A]
2022 Paper: ByteGraph: a high-performance distributed graph database in ByteDance (VLDB2022)
2024 Paper: Optimizing the Number of Clusters for Billion-Scalef Quantization-Based Nearest Neighbor Search (ICDE2024)

Professional Skills

- ▶ Experienced in C/C++ for many years, familiar with major distributed algorithms, and has participated in the design and implementation of distributed storage systems. Has a solid understanding of relational database theory and NoSQL.
- ▶ Knowledgeable in the basic algorithms and architecture of recommendation systems, and has implemented upstream and downstream services for recommendation systems.
- ▶ Familiar with the basic algorithms and architecture of precision advertising systems, understands the principles of search engines, and has contributed to the development of full-text indexing and ad indexing systems.
- ▶ Well-versed in common deep learning algorithms, with knowledge and experience in recommendation and search models, as well as training and inference engines. Built a graph training platform from the ground up, supporting all GNN and GE scenario training within the company.

Project Experiences at Work

- 2022.10 - Present **ByteDance - Infrastructure**
Responsible for building the graph training engine and platform in the ByteGraph team, including business integration, platform architecture design, and maintenance. Built the graph training platform and framework from the ground up, deeply involved in training task integration and algorithm optimization, supporting all graph training scenarios within the company.
- 2019.9 - 2022.9 **ByteDance - Search**
Responsible for ByteDance's general search data flow and the recall architecture for TikTok search, supporting model training, offline data statistics, sample backtracking, as well as database creation and recall architecture for TikTok search.

- 2017.12 - 2019.9 **ByteDance - Infrastructure**
Led ByteDance's DRC data synchronization service, supporting M (Musical.ly) and T (TikTok) integration and multi-data center, cross-DC data synchronization.
- 2016.11 - 2017.11 **Huawei Distributed Database**
Implemented distributed transactions based on PaxosCommit on InnoDB. Participated in implementation of Huawei's distributed database, working with an architecture similar to AWS Aurora.
- 2015.10 - 2016.11 **Meituan Cloud Self-Developed Object Storage System (S3Plus)**
Core designer and developer, responsible for the design and implementation of the RPC framework, participated in the full implementation of the StoreServer and some modules of the ProxyServer. Familiar with the entire system's design principles, implemented in C and Go.
- 2014.2 - 2014.11 **Alibaba Distributed Database Oceanbase 1.0 Version**
Participated in the design and implementation of the transaction module for the 1.0 cloud version, including distributed transaction design (two-phase commit) and log synchronization (based on Paxos). Designed the test framework for the leader election algorithm. Implemented in C++.
- 2015.6 - 2015.9 **Sohu Precision Advertising Index System**
Maintained and refactored the indexing system for precision advertising, implemented similarly to Lucene, providing search for ad candidate sets that meet specific targeting conditions and persisting the index to disk. Implemented in C++ in a distributed environment.